

BRAC University

CSE331 : Automata And Computability

Spring 2025

Quiz 1

SET - A

SOLVE

Marks :20

Duration: 40 mins

CO1	Name:	ID:	Sec:
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1. Two languages are defined as follows over the alphabets $\Sigma = \{0, 1\}$:

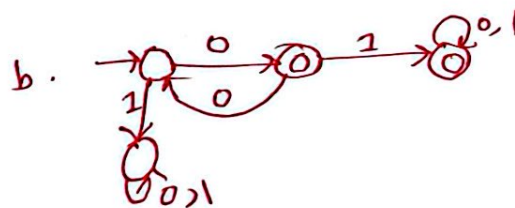
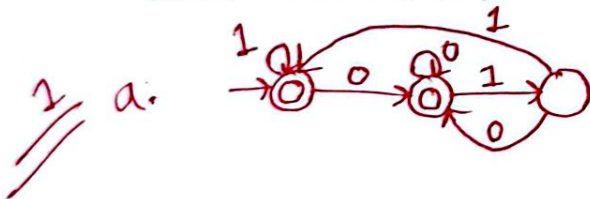
$L_1 = \{w \mid \text{doesn't end with } 01\}$

$L_2 = \{w \mid w \text{ starts with odd number of } 0\text{'s}\}$

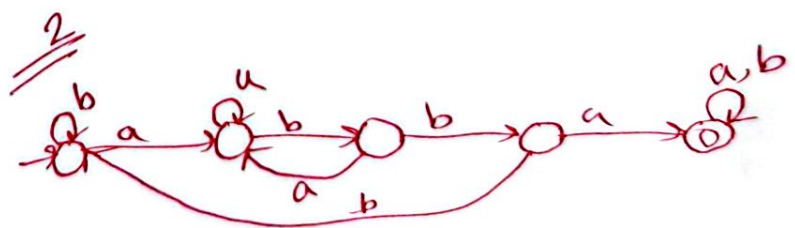
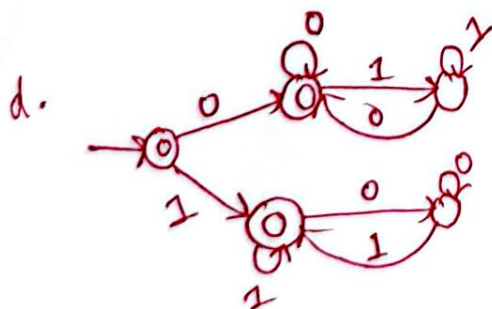
$L_3 = \{w \mid \text{the number of '01' substrings is equal to the number of '10' substrings in } w\}$

- Give a state diagram for that DFA that recognizes L_1 [4]
- Give a state diagram for that DFA that recognizes L_2 [4]
- If you were to use the "cross product" construction shown in class to obtain $L_1 \cup L_2$, how many states would it have? [3]
- Give a state diagram for that DFA that recognizes L_3 [3]

2. Design a DFA for the language $L = \{w \mid w \text{ contains "abba" as a substring}\}$ over the alphabets $\Sigma = \{a, b\}$ [6]



c. $3 \times 4 = 12$ states



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SET - B

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CO1	Name:	ID:	Sec:
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1. Two languages are defined as follows over the alphabets $\Sigma = \{0, 1\}$:

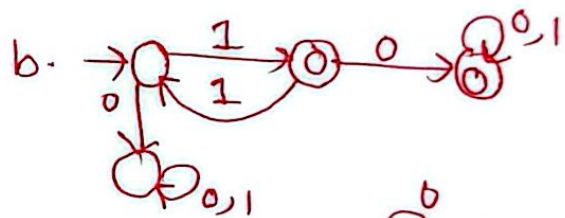
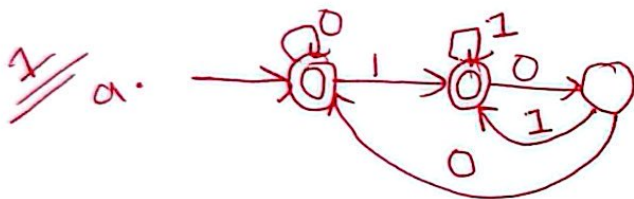
$L_1 = \{w \mid \text{doesn't end with } 10\}$

$L_2 = \{w \mid w \text{ starts with odd number of } 1\text{'s}\}$

$L_3 = \{w \mid \text{the number of '01' substrings is not equal to the number of '10' substrings in } w\}$

- Give a state diagram for that DFA that recognizes L_1 [4]
- Give a state diagram for that DFA that recognizes L_2 [4]
- If you were to use the "cross product" construction shown in class to obtain $L_1 \cup L_2$, how many states would it have? [3]
- Give a state diagram for that DFA that recognizes L_3 [3]

2. Design a DFA for the language $L = \{w \mid w \text{ contains "baab" as a substring}\}$ over the alphabets $\Sigma = \{a, b\}$ [6]



c. $3 \times 4 = 12$ states

